Integrated Sensing and Communication using GNU Radio and USRP

Internship Project Report

Submitted by:

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Abstract

Integrated Sensing and Communication (ISAC) is a key technology for future wireless systems, enabling the simultaneous use of the same waveform for both data transmission and environmental sensing. This report presents the implementation of a simple ISAC prototype using GNU Radio and USRP hardware. Two separate flowgraphs were designed: one for manual BPSK-based bit transmission and reception, and another for audio (song) transmission using WBFM. In both cases, real-time sensing was integrated by analyzing the received signal power. Experiments were conducted to detect metal and non-metal objects and to observe movement through power variation. The work validates the feasibility of low-cost ISAC systems using software-defined radios and provides a foundation for further exploration.

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Chapter 1

Introduction

1.1 Background

With the evolution towards 6G and intelligent wireless systems, the integration of sensing and communication has become a crucial research area. ISAC allows a single radio waveform to serve both data transmission and environmental sensing functions, improving spectrum utilization and enabling new applications such as radar-communication co-design, localization, and object detection.

1.2 Objective

The objective of this internship project is to:

- Implement and demonstrate the concept of Integrated Sensing and Communication (ISAC) using GNU Radio and USRP.
- Design two separate flowgraphs:
  - **Flowgraph 1:** Manual BPSK-based bit transmission and reception with integrated sensing.
  - **Flowgraph 2:** Audio (song) transmission using WBFM with sensing capability.
- Integrate real-time sensing by analyzing received signal power to detect environmental changes.
- Perform experiments for **metal vs. non-metal detection** and **movement detection**.
- Validate the feasibility of ISAC on a software-defined radio platform and explore its potential for future wireless systems.

Chapter 2

Tools and Setup

2.1 Software

- **GNU Radio:** Open-source signal processing framework used for flowgraph design.
- **Python Blocks:** Used for power averaging.

2.2 Hardware

- **USRP:** Universal Software Radio Peripheral for RF transmission and reception.
- **Antenna:** Omnidirectional antenna and rectangular wide band antenna

2.3 Parameters

- Center Frequency: 3.5 GHz
- Sampling Rate: 64 Ks/s for bit transmission and 48 Ks/s for audio streaming.
- Gain: Configured as per environment

Chapter 3

Methodology

3.1 Flowgraph 1: Bit Transmission using Manual BPSK with Sensing

3.1.1 Communication

1. **Bit Input:** A Vector Source is used to input user-defined bit sequences.
2. **Manual Mapping:** Bits are mapped manually as $0 \rightarrow -1$, $1 \rightarrow +1$.
3. **Carrier Modulation:** The baseband BPSK symbols are multiplied with a cosine carrier signal to perform BPSK modulation manually. This shifts the baseband signal to the desired carrier frequency for transmission.
4. **Transmission:** The modulated signal is transmitted using the USRP Sink with the configured center frequency, sampling rate (64 kS/s), and gain.
5. **Reception:** The received waveform is collected via USRP Source.
6. **Demodulation:** The received signal is first normalized using AGC2 and AGC3 to compensate for amplitude variations. It is then processed through a Costas Loop to recover the carrier phase. A Moving Average block smooths the signal, followed by a Low Pass Filter to remove high-frequency noise. The resulting signal is fed into a Binary Slicer to recover the transmitted bits (positive $\rightarrow$ 1, negative $\rightarrow$ 0).
7. **Verification:** The received bits are compared with transmitted bits for validation.

3.1.2 Sensing

1. The received signal is squared and averaged to compute instantaneous power.
2. A moving average filter smooths the power fluctuations.

3. Variations in power are mapped to signal strength.

4. Different objects (metal / non-metal) and movements are introduced near the antenna.

5. Observed changes in power level indicate presence, movement, or type of object.

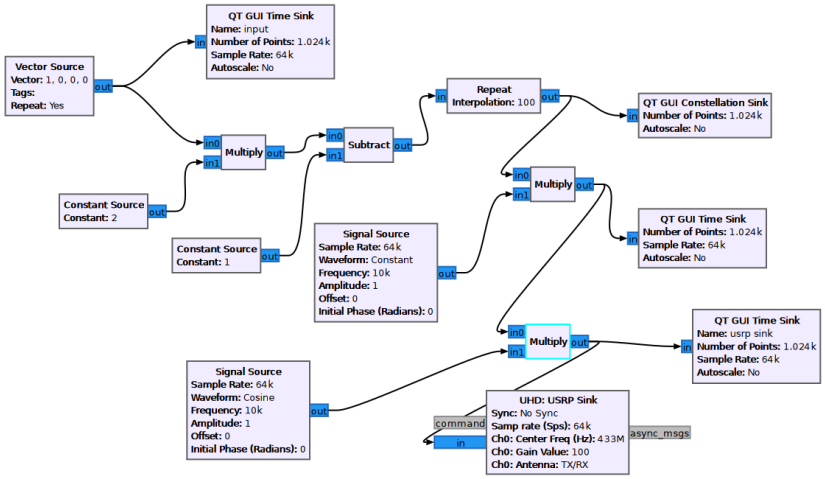


Figure 3.1: Flowgraph for Bit Transmission using Manual BPSK

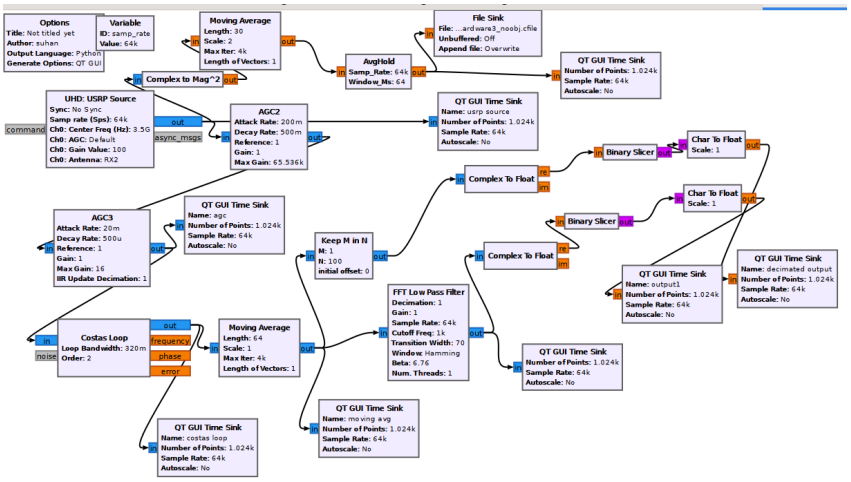


Figure 3.2: Flowgraph for Bit Reception and Sensing

3.2 Flowgraph 2: Song Streaming with Sensing

3.2.1 Communication

1. Audio samples are read from a WAV Source block.

2. The audio signal is modulated using a WBFM Transmit block.

3. The modulated signal is resampled using a Rational Resampler to match the USRP sampling rate.

4. The resampled audio signal is transmitted using USRP Sink.

5. At the receiver, the signal is collected via USRP Source.

6. The received signal is resampled as needed and demodulated using WBFM Receive to recover the audio.

3.2.2 Sensing

1. In the transmitter, a 1 kHz cosine wave of amplitude 0.1 is added to the resampled audio signal before transmission via USRP Sink.

2. At the receiver, a Band Pass Filter is used to extract the 1 kHz cosine wave.

3. The extracted signal is processed to compute the Received Signal Strength (RSS) for sensing.

4. Variations in RSS are mapped to object presence, type (metal/non-metal), or movement near the antenna.

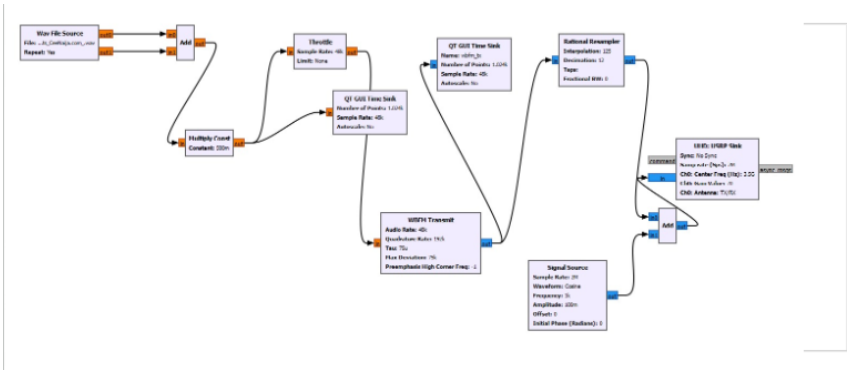


Figure 3.3: Flowgraph for song transmission and an added cosine wave for sensing

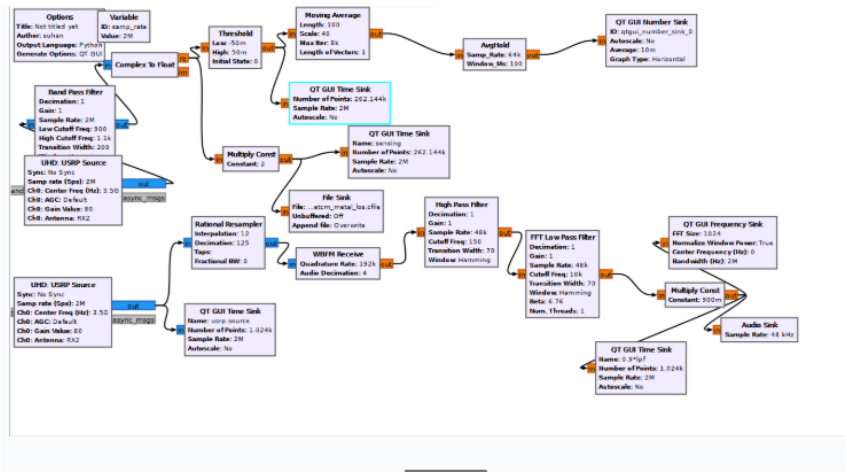


Figure 3.4: Flowgraph for song reception and sensing

Chapter 4

Experimental Results

4.1 Flowgraph 1: Bit Transmission and Sensing

1. **Bit Communication:** The system successfully transmitted user-defined bit sequences using manual BPSK mapping. Demodulated bits matched the transmitted sequence with negligible errors under stable channel conditions.

2. **Sensing via Power Measurement:** The received signal was squared and averaged to compute instantaneous power. A moving average filter smoothed the fluctuations, and variations in power were used to detect objects or movement near the antenna.

3. **Metal vs Non-Metal Detection:** Signal strength increased noticeably in the presence of metal due to reflection and absorption characteristics.

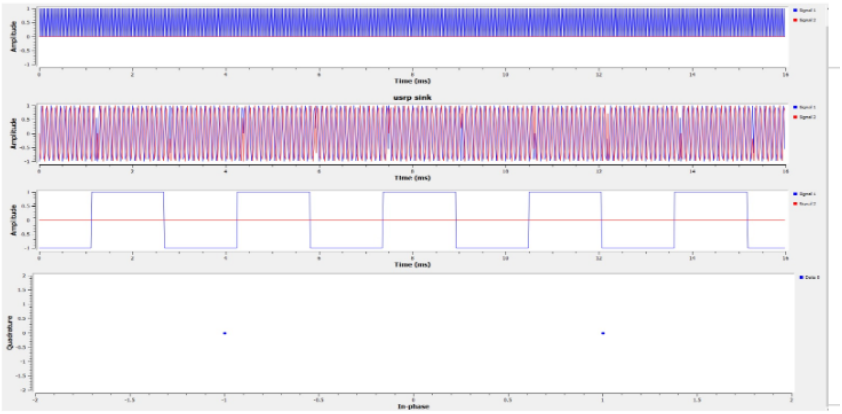


Figure 4.1: Transmitted bits, BPSK Modulated Signal, Resampled Bit transmission for better visualisation and Constellation Diagram respectively

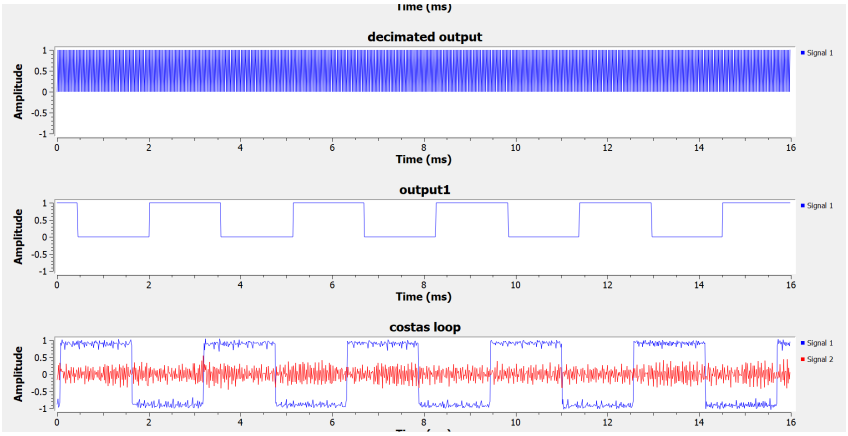
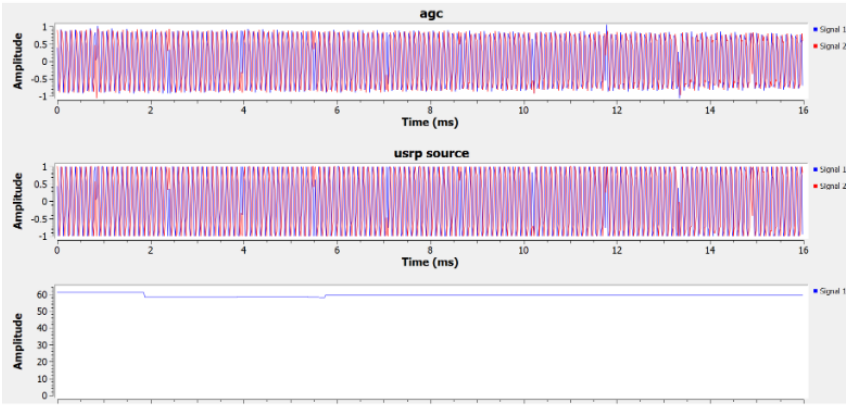


Figure 4.2: Received bits, Resampled version of received bits for better visualisation and Costas Loop Output respectively



Power signal

Figure 4.3: AGC output, Received BPSK Modulated Wave Output, Power Value respectively

4.2 Flowgraph 2: Song Streaming and Sensing

1. **Song Communication:** Audio samples from a WAV file were transmitted using WBFM modulation and a Rational Resampler. At the receiver, the signal was resampled and demodulated using WBFM Receive to recover the audio successfully.

2. **Sensing via Cosine Wave:** A 1 kHz cosine wave of amplitude 0.1 was added to the transmitted signal. At the receiver, a Band Pass Filter extracted this cosine, and the Received Signal Strength (RSS) was monitored to detect objects or movement.

3. **Metal vs Non-Metal Detection:** Signal strength increased noticeably in presence of metal due to reflection and absorption characteristics. Movement detection was confirmed by observing dynamic RSS variations over time as objects moved within the sensing zone.

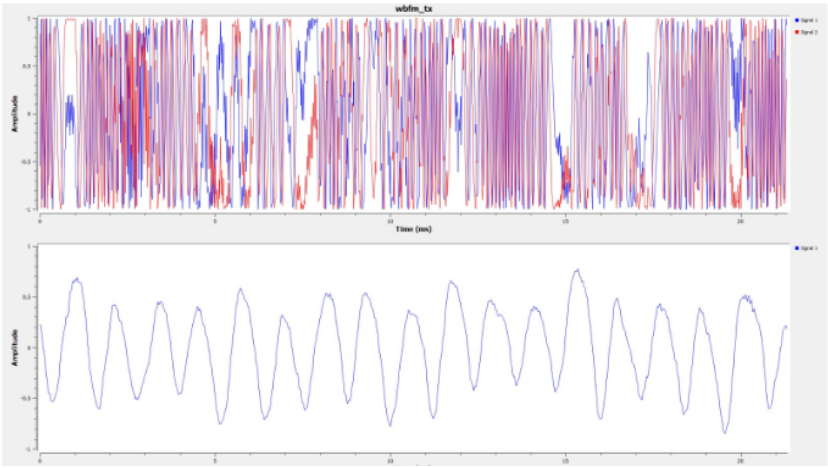


Figure 4.4: WBFM Modulated Signal, Audio Signal respectively

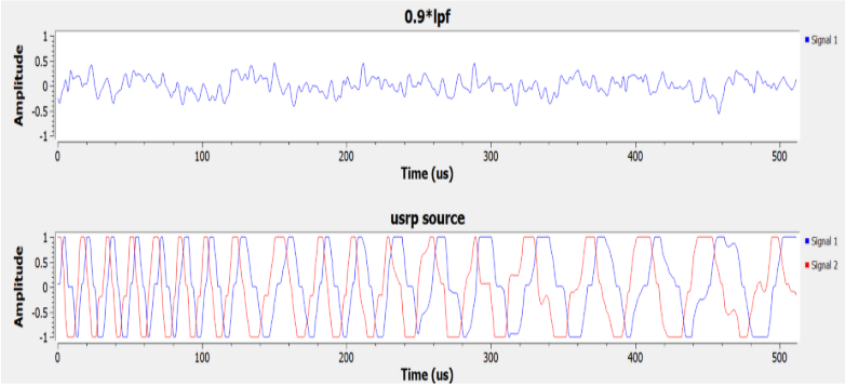


Figure 4.5: Received audio signal and Modulated wave respectively

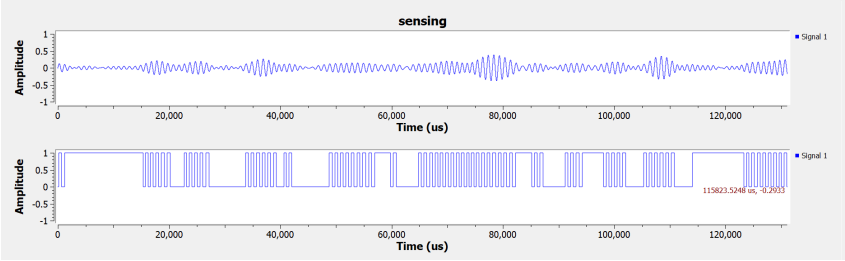


Figure 4.6: Extracted Cosine Wave for Sensing and Threshold Detector respectively

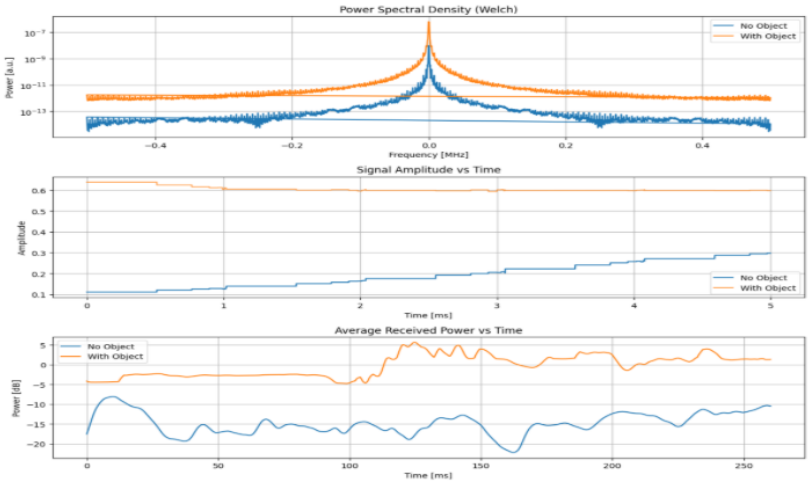


Figure 4.7: Difference visualisation (Object vs No Object)

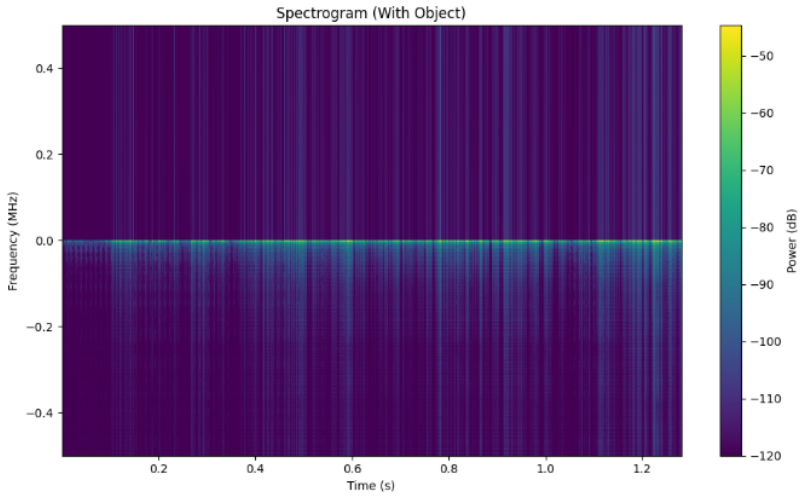


Figure 4.8: Spectrogram with Object

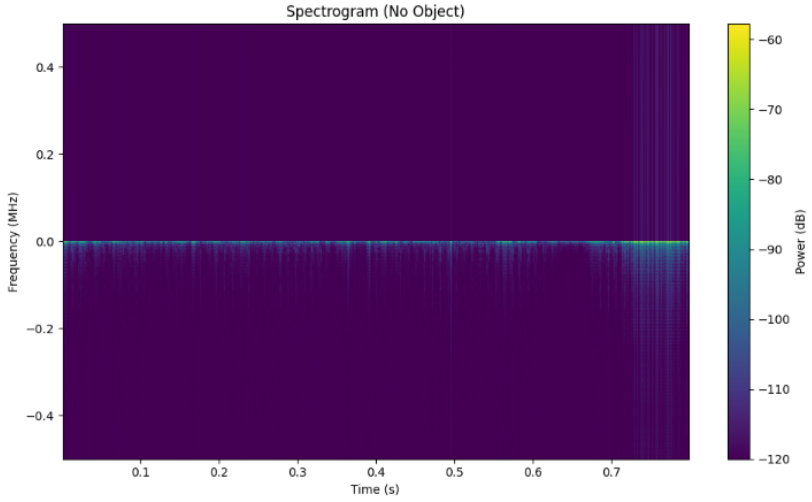


Figure 4.9: Spectrogram without Object

Chapter 5

Conclusion

The project successfully demonstrated a simple ISAC prototype using GNU Radio and USRP. Communication was implemented using manual BPSK for bit sequences and WBFM for audio streaming. Real-time sensing was achieved by monitoring received signal power and added waves, enabling detection of object presence, movement, and differentiation between metal and non-metal objects. These results confirm the feasibility of integrating sensing and communication functionalities within a software-defined radio platform.

5.1 Future Scope

- Extend to higher-order modulations (QPSK, QAM).
- Implement machine learning-based classification of objects.
- Combine with video or data communication for richer ISAC demonstrations.

Acknowledgements

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References

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3. S. Zhou, X. Liang, Y. Yu and H. Liu, "Joint Radar-Communications Co-Use Waveform Design Using Optimized Phase Perturbation," in IEEE Transactions on Aerospace and Electronic Systems, vol. 55, no. 3, pp. 1227-1240, June 2019, doi: 10.1109/TAES.2019.290925 keywords: Radar;Perturbation methods;Binary phase shift keying;Bit rate;Peak to average power ratio;Coexistent;co-use waveform design;co-use waveform design;joint radar-communications;passive radar;phase perturbation;RF convergence;spectral sharing;wireless communication.,